DR. THEJUS MAHAJAN

Biomedical Data Platforms · Interactive Data Tools · Scientific Software

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PROFILE

Physicist (PhD) building software for biomedical data, from backend to user interface. Available immediately; open to relocation anywhere in Germany.

- **Interactive data tools:** a Shiny tool for validating clinical registry data (6 modules, German/English); a web application with a FastAPI (REST) backend and a React/TypeScript frontend.
- **Biomedical data and metadata:** registry pipeline in R (143,000+ records, ~300 clinics); 241 correction rules and 16 success criteria moved out of the code into configuration files, output byte-identical.
- **Experimental data and validation:** calibration and quality control of collision data resolved by mass-to-charge ratio on a multi-detector setup (PhD); a simulation validated against measurements from revived strains (postdoc).

CLINICAL DATA AND DATA TOOLS

Biostatistics and Clinical Data Management (internship)

HealthTwiSt GmbH, under Dr. Andreas Busjahn

02/2026 - 04/2026 Berlin, Germany

Service company for the statistical analysis of clinical, genetic and scientific studies.

- Refactored the production R pipeline of the DeGIR[1] quality registry for interventional radiology (143,000+ records, ~300 clinics) from 1,834 to 1,349 lines; output verified byte-identical with `identical()`.
- Built on my own initiative an interactive Shiny tool for validating the registry data (6 modules, German/English).
- Moved 241 data-correction rules and 16 success criteria out of the code into configuration files. Found and documented two existing errors.
- Checked patient-level records; defined clinical endpoints (success / partial success / failure) for 26 intervention types.
- Extracted 9 functions into the R package `degirtools` (roxygen2, `devtools::check()` with 0 errors).

Web Application and Interpretability (own project)

github.com/thejusmahajan/lc0-interpretability

07/2026 - present Hamburg, Germany

- Built a chess-coaching web application: FastAPI (REST) backend, React/TypeScript frontend; Python/PyTorch analysis of what the Leela Chess Zero network attends to.
- Automated test suite (pytest) for the backend.
- Found two silent errors in my own analysis, one after publication; corrected both publicly.

Continuing Education: Applied Bioinformatics & Biostatistics

CQ Beratung + Bildung GmbH, full-time

08/2025 - 02/2026 Berlin, Germany

Completed 04/2026 (with the internship). Module grades 1.0 – 2.1.

- Applied Biostatistics in R, grade 1.0: linear, logistic and mixed models, PCA, discriminant and cluster analysis, MCMC, random forests and boosting.
- Bioinformatics & Databases, SQL/MySQL and relational database design (1.4); Programming in Python (1.2); final project: a Nextflow (DSL2) pipeline.

SKILLS

Web & APIs
FastAPI, REST APIs, React, TypeScript, JavaScript, R Shiny

Data
SQL/MySQL, relational database design, configuration-driven metadata, de-identification in R, FAIR and research data management (certified, internship)

Programming
Python (NumPy, Pandas, PyTorch, JAX, PySpark, scikit-learn), R (tidyverse, tidymodels, Shiny), Fortran, Bash

Statistics & ML
Linear, logistic and mixed models, multivariate methods, MCMC, random forests (grade 1.0); PyTorch, Spark MLlib; LLM agents with tool calls (MCP), checked against SQL ground truth

Computing
GPUs (V100), PyTorch multi-GPU (DDP), PBS batch jobs (HLRS), OpenMP, Linux cluster (SLURM, MPI)

Code quality
Git, automated tests, regression checks against reference output, R packages and documentation (roxygen2, devtools), Quarto

HPC AND AI TRAINING

Deployable Data Analysis & AI Pipelines with HPC
Supercomputing-Akademie, HLRS Stuttgart, 09–10/2026, 50 h. Exam passed; certificate issued 05.11.2026.

- PBS batch and interactive jobs on GPU (V100) and CPU (Skylake) nodes; multi-GPU training with PyTorch DDP on 4 GPUs.
- Compared two open-weight LLMs as tool-calling agents, with answers checked against SQL reference queries; traced an invented figure to the agent step that produced it.

Parallel Programming with MPI
HLRS Stuttgart, online. In progress, 02.11.–04.12.2026.

IBM: Deep Learning with PyTorch
Coursera, completed 08/2026.

Introduction to Supercomputing
Jülich Supercomputing Centre, 11/2025.

[1] DeGIR: Deutsche Gesellschaft für Interventionelle Radiologie und minimal-invasive Therapie.

MODELLING RESEARCH

Postdoctoral Scientist, Marine Ecosystem Modelling
Universität Hamburg, Institute of Marine Ecosystem and Fishery Science
08/2021 - 01/2025 Hamburg, Germany

- Built a 0-D Lagrangian individual-based model (IBM-CLC) of *Nodularia spumigena* with a temperature optimum (T_{opt}) per cell, thermal performance curves and akinete dormancy. Manuscript in final preparation.
- Reproduced the measured T_{opt} of strains revived from sediment (Medwed et al., 2024) within the error bars.
- Traced a shift of about 1 °C in predicted T_{opt} to the random-number stream; fixed with standing genetic variation in germinating cells and a large akinete pool.
- Parallelised the particle loops of the Fortran model with OpenMP; ported the simulation engine to Python/JAX for GPUs, output binary-compatible with the Fortran format. Linux cluster (SLURM, MPI); NetCDF/HDF5.

Parental leave 01/2024 – 12/2024.

Guest Scientist
Helmholtz-Zentrum Hereon, Ecosystem Modelling
05/2025 - 10/2025 Geesthacht, Germany

Finalised the modelling framework of the IBM-CLC model and prepared the simulation data for its manuscript.

TEACHING AND USER MATERIAL

- Taught on the BSc course Marine Ecosystem Modelling, Universität Hamburg (2022–23); MSc guest lecture “From 0D to 1D”.
- HealthTwiSt: wrote an introduction to tidymodels as a Quarto document; moved the teaching scripts from caret to tidymodels.
- 2.5 years of one-to-one teaching in physics and chess (2018–2021).

LANGUAGES

English ●●●●●
Fluent (C1)

German ●●●○○
B1 (Goethe-Zertifikat), B2 exam booked for 10.12.2026

French ●●○○○
Basic

Malayalam ●●●●●
Native

Tamil, Hindi ●●●○○
Working proficiency

EARLIER CAREER

Doctoral Researcher
Université Paris-Saclay, ISMO
10/2015 - 09/2018 Orsay, France

- Collision experiments on the AGAT setup at the Orsay Tandem accelerator: fragments separated by mass-to-charge ratio with electrostatic deflection and detected in coincidence, event by event; data acquisition, calibration and quality control of the multi-detector data.
- Classical Trajectory Monte Carlo (CTMC) calculations at Universidad Autónoma de Madrid and Université de Bordeaux; branching ratios contributed to the KIDA astrochemistry database.

Physics Tutor & Chess Coach
Freelance / Tutorwaves Solutions Inc.
10/2018 - 03/2021 Kerala, India

One-to-one teaching in physics and chess.

Relocation to Germany
04/2021 - 07/2021 Hamburg, Germany

Move from India to Germany; job search.

EDUCATION

Ph.D. in Lasers, Molecules and Atmospheric Radiation
Université Paris-Saclay
10/2015 - 09/2018 Orsay, France

Thesis applied to astrochemistry: *Excitation and fragmentation of C_nN^+ (n=1–3) in collisions with He atoms*.

M.Sc. in Physics
National Institute of Technology Calicut
07/2012 - 12/2014 Calicut, India

First Class with Distinction (CGPA 8.71/10). B.Sc. in Physics, University of Calicut, 2009–2012.

PUBLICATIONS

Four peer-reviewed journal articles (*Physical Review A*; *J. Phys. B: At. Mol. Opt. Phys.*; *Astronomy & Astrophysics*; *Molecular Astrophysics*) and two refereed conference papers (*J. Phys.: Conf. Ser.*), 2017–2020; first author on three. Three conference presentations. One modelling manuscript (IBM-CLC) in final preparation.

REFEREES

Available on request.